

BAS112

Discrete Mathematics & Linear Algebra

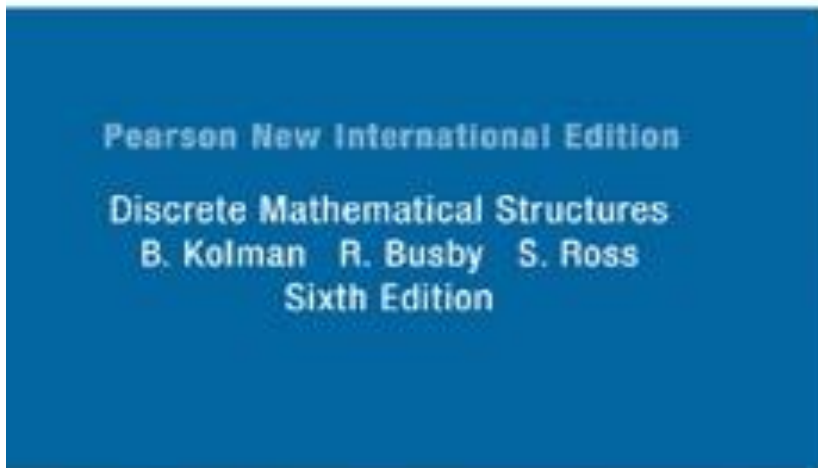
Lecture 1

Mathematical Logic

Chapter 2 – Part 1

Chapter 2

Logic



Chapter 2

The Foundations: Logic and Proofs

Propositions: A **proposition** is a declarative sentence (that is, a sentence that declares a fact) that is either **true or false**, but not both.

Example(1)

All the following declarative sentences are propositions .

1. Alexandria is the capital of Egypt

2. $1 + 1 = 2$

3. $2 + 2 = 3$

4. Washington , D.C., is the capital of the United State of America.

Propositions 2 and 4 are true; whereas 1 and 3 are false.

Some sentences that are not propositions are given in the following example

Example (2)

Consider the following sentences.

1. What time is it?
2. Read this carefully.
3. $x+1 = 2$ **This is called a predicate because if the value of the unknown x is been specified then at this moment it will be a proposition**
4. $x + y = z$

Sentences 1 and 2 are not propositions because they are not declarative sentences. Sentences 3 and 4 are not propositions because they are neither true nor false.

Note that : each of sentences 3 and 4 can be turned into a proposition if we assign values to the variables.

We use letters $p, q, r, s \dots$ to denote **propositions**.

The **truth value** of a proposition is **true**, denoted by **T**, if it is a true proposition and

The **truth value** of a proposition is **false**, denoted by **F**, if it is a false proposition.

Many mathematical statements are constructed by combining one or more propositions.

New propositions, called **compound propositions**, are formed from existing propositions using logical operators.

Logical Operators

- Many proposition are constructed by combining one or more propositions using logical operators (“ **not**” , “ **and**” , “**or**” , “ **implies**” and “ **if and only if**”) to form compound propositions . To determine the truth value of a compound proposition we need to know the effect of logical operators as well as the truth values of the constituent propositions. This can be summarized using a truth table.

The basic logical connectives are listed in the following table

Connective	Symbol
and	$\wedge$
or	$\vee$
if - then	$\rightarrow$
if and only if	$\leftrightarrow$
not	$\neg$

Definition (1): Negation (Not) $\neg p$

Let p be a proposition. The **negation of p** , denoted by $\neg p$ (also denoted by $\bar{p}$), is the statement “It is not the case that p . “

The proposition is read “not p “.

Table 1 displays the truth table for the negation of a proposition p .

p	$\neg p$
T	F
F	T

We will now introduce the logical operators that are used to form new propositions from two or more existing propositions. The logical operators are also called **connectives**.

Definition (2): And ($p \wedge q$)

Let p and q be propositions. The **conjunction** of p and q , denoted by $p \wedge q$ is the proposition “ p and q ”.

The conjunction is **true** when **both p and q** are **true** and is **false otherwise**.

Table 2 displays the truth table of $p \wedge q$.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Table 2 : The connective “and”

Example (3): Find the conjunction of the propositions p and q where p is the proposition “Today is Friday “ and q is the proposition “It is raining today. “

Solution: The conjunction of these propositions, $p \wedge q$ is the propositions

“Today is Friday and it is raining today “

This proposition is true on rainy Fridays and is false on any day that is not a Friday and on Fridays when it does not rain.

Example (4): The proposition
“Paris in France and $2+2 = 5$ “ is false proposition,
since it a compound proposition from the proposition
“**p = Paris in France**” and “**q = $2+2 = 5$** ”
p is true, while q is false, then, $p \wedge q$ is false.

Example (5) : Using the truth table , prove the following

$$(a) \quad p \wedge q \equiv q \wedge p \quad (\text{Commutative Law})$$

$$(b) \quad p \wedge p \equiv p \quad (\text{Idempotent Law})$$

$$(c) \quad p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r \quad (\text{Associative Law})$$

Solution : By using the following table we can prove (c) (columns 7 & 8) and proof of (a) (columns 4 &5)

1	2	3	4	5	6	7	8
p	q	r	$p \wedge q$	$q \wedge p$	$q \wedge r$	$p \wedge (q \wedge r)$	$(p \wedge q) \wedge r$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	F	F
T	F	T	F	F	F	F	F
T	F	F	F	F	F	F	F
F	T	T	F	F	T	F	F
F	T	F	F	F	F	F	F
F	F	T	F	F	F	F	F
F	F	F	F	F	F	F	F

(b)

$$p \wedge p \equiv p$$

p	p	$p \wedge p$
T	T	T
F	F	F

Definition 3: Or $(p \vee q)$

Let p and q be propositions.

The **disjunction** of p and q , denoted by $p \vee q$, is the proposition “ p or q .”

The disjunction $p \vee q$ is **false** when both p and q are **false** and is true otherwise .

Table 3 displays the truth table for **$p \vee q$** .

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Table 3 : The connective “ or”

Example (6): What is the disjunction of the propositions p and q where p and q are the same propositions as in Example (3)?

Solution: The disjunction of p and q , $P \vee Q$, is the proposition

” Today is Friday or it is raining today “

This proposition is true on any day that is either a Friday or a rainy day (including rainy Fridays).

It is only false on days that are not Fridays when it also does not rain.

Example (7) : Using the truth table , prove the following

(a) $\mathbf{p \vee q \equiv q \vee p}$ **(Commutative Law)**

(b) $\mathbf{p \vee (q \vee r) \equiv (p \vee q) \vee r}$ **(Associative Law)**

(c) $\mathbf{p \vee p \equiv p}$ **(Idempotent Law)**

Solution : By using the following table we can prove
 (b) (columns 7 & 8) and proof of (a)
 (columns 4 & 5)

p	q	r	$p \vee q$	$q \vee p$	$q \vee r$	$p \vee (q \vee r)$	$(p \vee q) \vee r$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	T	T	T	T	T
T	F	F	T	T	F	T	T
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	F	F	T	T	T
F	F	F	F	F	F	F	F

(c) $p \vee p \equiv p$

p	p	$p \vee p$
T	T	T
F	F	F

Definition 4: Conditional Statements ($p \rightarrow q$)

We will discuss several other important ways in which propositions can be combined.

Definition 4: Let p and q be propositions.

The conditional statement $p \rightarrow q$ is the proposition

“ if p then q . “

The conditional statement $p \rightarrow q$ is **false when p is true and q is false**, and true otherwise.

In the conditional statement $p \rightarrow q$, p is called the hypothesis and q is called the conclusion.

The statement $p \rightarrow q$ is called a **conditional statement** because $p \rightarrow q$ asserts that q is true on the condition that p holds. A conditional statement is also called an **implication**.

Table 4 displays the truth table for $p \rightarrow q$

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Note that : the statement $p \rightarrow q$ is true when both p and q are true and when p is false (no matter what truth value q has)

Example (8) :

- (i) The proposition “ if $2b \neq b$, where b is a positive number , then $2 \neq 1$ “ is **true proposition**
- (ii) The proposition “ if $2b \neq b$, where b is a positive number , then $2 = 1$ “ is **false proposition**
- (iii) The proposition “if $2b = b$, where b is a positive number, then $4 = 4$ “is **true proposition**
- (iv) The proposition “if $2b = b$, where b is a positive number, then $2 = 1$ “is **true proposition**

Example (9) : Construct truth tables for

(a) $p \rightarrow \neg(q \vee p)$

(b) $(p \vee \neg q) \rightarrow (p \wedge q)$

Solution :

(a)

p	q	$q \vee p$	$\neg(q \vee p)$	$p \rightarrow \neg(q \vee p)$
T	T	T	F	F
T	F	T	F	F
F	T	T	F	T
F	F	F	T	T

(b)

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \rightarrow (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Example (10) : Using the truth table , prove the following

(a) $\mathbf{p \rightarrow q \equiv \neg(p \wedge \neg q)}$

(b) $\mathbf{p \rightarrow q \equiv \neg p \vee q}$

Solution:

(a)

p	q	$\neg q$	$p \wedge \neg q$	$\neg(p \wedge \neg q)$	$p \rightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	F	T	T
F	F	T	F	T	T

The columns corresponding to $\neg(\mathbf{p} \wedge \neg \mathbf{q})$ and $\mathbf{p} \rightarrow \mathbf{q}$ are identical, thus these two compound propositions are logically equivalent

(b)

$\mathbf{p}$	$\mathbf{q}$	$\neg \mathbf{p}$	$\neg \mathbf{p} \vee \mathbf{q}$	$\mathbf{p} \rightarrow \mathbf{q}$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

The columns corresponding to $\neg \mathbf{p} \vee \mathbf{q}$ and $\mathbf{p} \rightarrow \mathbf{q}$ are identical, thus these two compound propositions are logically equivalent

Definitions:

(1) The converse of $p \rightarrow q$ is $q \rightarrow p$

(2) The inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$

(3) The contrapositive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$

Example(11): If we consider the statement
If **x is divisible by 4** then **x is an even number**

(1) The converse is:

If **x is an even number** then **x is divisible by 4**

(2) The inverse is:

If **x is not divisible by 4** then **x is not an even number**

(3) The contrapositive is :

If **x is not an even number** then **x is not divisible by 4**

Note That :

- (i)** Conditional statement and its converse are not equivalent
- (ii)** Conditional statement and its inverse are not equivalent
- (iii)** A conditional statement is equivalent to its contrapositive
- (iv)** The converse and the inverse of a conditional statement are equivalent to each other

Example (12) : Give the converse and the contrapositive of the implication “If it is raining, then I get wet.”

Solution:

We have p : It is raining; and q : I get wet.

(i) The converse is :

$q \rightarrow p$: If I get wet, then it is raining.

(ii) The contrapositive is:

$\neg q \rightarrow \neg p$: If I do not get wet, then it is not raining.

Definition 5: Biconditional Statements ($p \leftrightarrow q$):

We now introduce another way to combine propositions that expresses that two propositions have the same truth value.

Definition 5: Let p and q be propositions.

The biconditional statement $p \leftrightarrow q$ is the proposition

“ p if and only if q . “

The biconditional statement $p \leftrightarrow q$ is **true** **when p and q have the same truth values** and is false otherwise.

Biconditional statements are also called bi-implications.

Table 5 displays the truth table for $p \leftrightarrow q$

P	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Note that :

(i) The statement $p \leftrightarrow q$ is **true** when both the conditional statements $p \rightarrow q$ and $q \rightarrow p$ are **true** and is false otherwise.

That is why we use the words “if and only if “to express this logical connective and why it is symbolically written by combining the symbols $\rightarrow$ and $\leftarrow$.

(ii) There are some common ways to express $p \leftrightarrow q$:

“p is necessary and sufficient for q “ “p iff q “ .

The last way of expressing the biconditional statement $p \leftrightarrow q$ uses the abbreviation “iff” for “if and only if”.

(iii) $p \leftrightarrow q$ has exactly the same truth values as $(p \rightarrow q) \wedge (q \rightarrow p)$.

(iv) p is a sufficient condition for q means
“if p then q ” ($p \rightarrow q$)

(v) p is a necessary condition for q means
“if not p then not q ” ($\neg p \rightarrow \neg q$)

Example (13) : Using the truth table , prove the following

$$p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p) \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

Solution:

p	q	$\neg p$	$\neg p \vee q$	$\neg q$	$\neg q \vee p$	$(\neg p \vee q) \wedge (\neg q \vee p)$	$p \leftrightarrow q$
T	T	F	T	F	T	T	T
T	F	F	F	T	T	F	F
F	T	T	T	F	F	F	F
F	F	T	T	T	T	T	T

$(\mathbf{p} \wedge \mathbf{q})$	$(\neg \mathbf{p} \wedge \neg \mathbf{q})$	$(\mathbf{p} \wedge \mathbf{q}) \vee (\neg \mathbf{p} \wedge \neg \mathbf{q})$
T	F	T
F	F	F
F	F	F
F	T	T

The columns corresponding to $(\neg \mathbf{p} \vee \mathbf{q}) \wedge (\neg \mathbf{q} \vee \mathbf{p})$, $\mathbf{p} \leftrightarrow \mathbf{q}$ and $(\mathbf{p} \wedge \mathbf{q}) \vee (\neg \mathbf{p} \wedge \neg \mathbf{q})$ are identical , thus these three compound propositions are logically equivalent

Example (14) : Construct the truth table ,for the following proposition:

$$\neg(p \wedge q) \rightarrow (\neg p \vee \neg q)$$

Solution:

p	q	$\neg p$	$\neg q$	$p \wedge q$	$\neg(p \wedge q)$	$\neg q \vee \neg p$	$\neg(p \wedge q) \rightarrow (\neg p \vee \neg q)$
T	T	F	F	T	F	F	T
T	F	F	T	F	T	T	T
F	T	T	F	F	T	T	T
F	F	T	T	F	T	T	T